

Task-1

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① HTML: HTML, which stands for Hyper Text Markup Language, is the standard language used to create and structure content on the World wide web. It acts as the skeleton or blueprint of a web page, telling web browsers how to display text, images, links, and Multimedia.

↳ World wide web (WWW) is a global system of connected webpages, pictures, videos & files. You reach these files through the internet by using links and a web browser.

↳ Web Browser - is a software application used to access the internet, while a Website - is a collection of web pages hosted on a server.

Eg: You use a browser (the tool) to open & view a Website (the destination).

↳ Web page - A web page is a single digital document viewed in a web browser. It holds text, pictures, and videos. It has a unique web address called a URL. Many web pages grouped together under one name make up a website.

Example:

Web page - A single page/document on the Internet

Eg: Amazon's product details page for a particular phone

Website - A collection of related Web pages under one domain.

Eg: Amazon.com - contains thousands/millions of pages such as products, art, orders etc.

Webbrowser - software used to open and view website

Eg: Google Chrome - the application you use to access amazon.

Simple: Website - House

Web page - one room in the house

Web browser - vehicle you use to reach the house.

So, chrome is not a website. It is a web browser used to access websites.

② CSS: CSS is a stylesheet language used to control the presentation, layout, and appearance of HTML elements on a web page.

→ CSS (Cascading Style Sheet) is a language used to style and design web pages. It controls things like:

- Colors
- Size & spacing
- Borders & Backgrounds
- Fonts and text
- Layout

Simple example: HTML creates a button, while CSS makes the button colorful, changes its size, and adjusts its appearance.

③ SQL: Structured Query Language is a language used to store, retrieve, update and manage data in databases.

Ex: Suppose we have a student's table

<u>ID</u>	<u>Name</u>	<u>Course</u>
1.	Abi	CSE
2.	Sangal	IT

Website

HTML

CSS

SQL

REACT

JS

Bootstrap

Tailwind CSS

Qns

To display all students, We use

`SELECT * FROM students;`

→ This command retrieves all the data from the student table.

Interview def: SQL is a standard language used to communicate with and manage data in relational database.

↳ Database: A database is an organized collection of data that can be stored, accessed, and managed easily.

Ex: A college database can store:

- Student names, IDs, courses, marks, contact details.

Think of it as a digital storage system for data.

↳ Relational Database: A relational database stores data in tables (rows and columns) and can create relationships between different tables.

Ex: Student table

student-ID	Name
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201	Abi
-----	-----

202	Sanjai
-----	--------

Course table

course-ID	student-ID	Course
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1	101	CSE
---	-----	-----

2	102	IT
---	-----	----

Here, student-ID connects the two tables, That's why it is called relational.

Easy way to remember:

→ Database = A place where data is stored.

→ Relational Database → Data is stored in related table

→ SQL - Language used to work with the data in relational database.

Examples of relational databases: MySQL, PostgreSQL, Oracle, Microsoft SQL server.

④ Javascript: JS is a programming language used to make web pages Interactive and dynamic

↓
the website responds to user's action.

↳ click submit button - website shows form submitted successfully.

↳ clicking a menu - menu opens.

dynamic - the content of webpage can change without manually editing the web page.

↳ ex: you select "Tamilnadu" from a dropdown - The page displays cities such as Chennai, Thanjavur, Tiruchy etc.

Simple: Interactive - Responds to user actions
Dynamic - Changes content/appearance.

Example of JS: Suppose you have a button:

Before clicking: click me.

↳ When the user clicks it, Javascript can display:

Hello! Welcome to my website.

Other examples of JS:

- Button click actions
- Form validation
- calculations
- updating page content without refreshing
- Adding products to a shopping cart.

Interview def of JS: Javascript is a programming language used to add interactivity and dynamic behavior to web page and Web Application.

↳ Web Application: - is a software application that you use through a web browser such as chrome, edge or firefox.

→ It allows user to perform task or interact with data rather than simply viewing.

Eg: Gmail - send & receive emails.

Google docs - create & edit documents.

online banking - check balance and transfer money.

Simple diff: Website: Mainly provides information.

eg: Company website showing its services.

Web application: Allows you to perform actions.

eg: An online banking application where you can transfer money.

⑤ React / React.js

Inter def: React JS is a javascript library developed by meta for building interactive and reusable user interfaces, especially for web applications.

Easy to rem React = React.js = Javascript library for building UI.

→ React is a Javascript library used to build the user interface (UI) of web applications.

→ It helps developers create reusable components such as buttons, menus, forms and cards.

Simple ex: Imagine an online shopping website!

↳ - Nav bar (Navigation bar) → one react component
- search bar - one react component.

product card - one react component

Cart - one react component

→ A product card could display: Phone ₹ 20000
Add to cart

- Instead of writing same code for every product
React allows you to create one product card
components and reuse it for many products.

Another Inter def: React is JS library used to
build interactive and reusable user interfaces for
web applications.

⑥ Bootstrap: Bootstrap is a CSS framework used to
create responsive and attractive website quickly.

→ It provides ready made styles and components

Such as:

- buttons
- Navigation bars
- Forms
- Cards
- Tables
- Alerts.

Simple ex: 1) Without bootstrap, you may need to write
CSS yourself to create a button.

- With bootstrap you can use a ready made class:

```
<button class="btn  
  btn-primary"> Login </button>
```

ii) You can create a navigation bar using bootstrap's
readymade components instead of designing the entire
navbar from scratch.

Why use Bootstrap? 1) Without Bootstrap:
you write more CSS yourself

ii) With Bootstrap: you use Bootstrap's pre-built
classes and components, so development is faster.

Interview def: Bootstrap is a popular front-end CSS framework that provides pre-built components and responsive layouts for developing websites quickly.

⑦ Tailwind CSS: Tailwind CSS is a CSS framework that helps you design websites using small, ready-made utility classes directly in your HTML.

Simple ex: Instead of writing separate CSS:

```
< button class = "bg-blue-500 text-white  
px-4 py-2 rounded">  
</button>
```

Here:

bg-blue-500 → background color

text-white → text color.

px-4 → horizontal padding

py-2 → vertical padding

rounded → rounded corner.

Bootstrap vs Tailwind CSS

↓
Bootstrap provides more pre-designed components

↳ provides small utility classes that you combine to create your own design.

Interview def: Tailwind CSS is a utility-first CSS framework used to build customized and responsive user interfaces quickly."

Overview

Topic

Simple meaning

- ① Web page - A single page on the 'internet'
- ② Website - A collection of related web pages
- ③ Webbrowser - Software used to access websites, eg., chrome.
- ④ HTML - Creates the structure of a webpage
- ⑤ CSS - Designs & styles the webpage
- ⑥ Javascript - Adds action and interactivity to a web page
- ⑦ React / React.js - Javascript library for building interactive user interfaces.
- ⑧ Bootstrap - CSS framework with ready made components and styles.
- ⑨ Tailwind CSS - CSS framework using utility classes for styling.
- ⑩ SQL - Language used to work with data in databases.
- ⑪ Database - Organized collection of data.
- ⑫ Relational Database - Database that stores data in related tables
- ⑬ Navbar - Navigation section that helps users move around a website
- ⑭ Web Application - Application accessed through a web browser to perform tasks

How they work together: For example, when building an online shopping web application:

HTML → Creates the product page structure
↓
CSS/Bootstrap/Tailwind CSS → Makes it look good
↓
JavaScript → Adds action such as clicking Add to cart
↓
React → Helps build the UI using reusable components.
↓
SQL + Database → Stores products, users, orders, etc.
↓
Web Browser → Allows the user to access the application.

One-line flow to remember

⇒ HTML = Structure, CSS = Design, JavaScript = Behavior,
React = UI, SQL = Data, Database = Storage, Browser = Access.

Product based company: Imagine a company creates its own shopping application and sells/operates it for millions of users. The company owns the product.

Service - based company: A company gets a project from a bank to develop a banking application. It builds the application according to the client's requirements.

Easy way to remember:

product-based → We build our own product.

service-based → We build solutions for clients.

product - Based company Vs service based company

	PB company	S-B company
Meaning	- Build and sells its own product/software	provides IT services to clients

Main focus	- Developing & improving its products	Completing client projects
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Customers	- usually many users/customers	usually other companies / clients
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Examples	- Google, Microsoft, Amazon	TCS, Infosys, Wipro
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Work	- product development, maintenance, new features	Client projects, development, testing, support
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Learning-oriented company → Focuses strongly on employee learning and development.
